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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

*Off-Campus Centre of Nitte (Deemed to be University)*

**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

## CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(For CS, CB)

Duration: 3 Hours — Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

**1. Explain what is meant by PRODUCT with reference to one of the eight principles as per the ACM/IEEE Code of Ethics.**

A) The product should be easy to use
B) Software engineers shall ensure that their products and related modifications meet the highest professional standards possible
C) Software engineers shall ensure that their products and related modifications satisfy the client
D) It means that the product designed/created should be easily available

**2. Which of the following property does not correspond to a good Software Requirements Specification (SRS) ?**

A) Verifiable
B) Ambiguous
C) Complete
D) Traceable

**3. Requirements should specify ______ but not 'how' aspects of a software.**

A) 'what'
B) 'how'
C) 'why'
D) 'which'

**4. Software Engineering is an engineering discipline that is concerned with?**

A) how computer systems work
B) theories and methods that underlie computers and software systems
C) all aspects of software production
D) all aspects of computer-based systems development, including hardware, software and process engineering

**5. The major shortcoming of waterfall model is?**

A) the difficulty in accommodating changes after requirement analysis
B) the difficult in accommodating changes after feasibility analysis
C) the system testing
D) the maintenance of system

**6. Project risk factor is considered in which SDLC model?**

A) Incremental Model
B) Waterfall model
C) Prototyping model
D) Spiral model

**7. If requirements are easily understandable and defined then which model is best suited?**

A) Spiral model
B) Waterfall model
C) Incremental Model
D) Prototyping model

8. **Identify which phase is not available in SDLC?**
   A) Coding B) Testing
   C) Maintenance D) Abstraction
9. **Which model in system modeling depicts the dynamic behavior of the system?**
   A) Context Model B) Behavioral Model
   C) Data Model D) Object Model
10. **_________ and ___________ diagrams of UML represent Interaction modeling.**
    A) Use Case, Sequence B) Class, Object
    C) Activity, State Chart D) Class, State
11. **Which of the following is a complementary approach to function-oriented approach?**
    A) Object oriented analysis B) Object oriented design
    C) Structured approach D) Both Object oriented analysis and design
12. **Sequence of messages is emphasized by which UML diagram?**
    A) state chart B) sequence
    C) activity D) collaboration
13. **Agile Software Development is based on?**
    A) Incremental Development B) Iterative Development
    C) Linear Development D) Both Incremental and Iterative Development
14. **According to Agile manifesto what carries more value?**
    A) Individuals and interactions over processes and tools
    B) Individuals and interactions over people and technique
    C) Individuals and interactions over projects and tools
    D) Individuals and interactions over products and tools
15. **Verifying that whether individual software components are functioning correctly and identifying the defects in them is objective of which level of testing?**
    A) Integration Testing B) Acceptance Testing
    C) Unit Testing D) System Testing
16. **_________ and_________ processes are concerned with checking that software being developed meets its specification and delivers the functionality expected by the people paying for the software.**
    A) Verification and Validation B) Analysis and Design
    C) Code and Design D) Analysis and Code
17. **If an organization is unsure of its cost estimate, it may increase its price by a contingency over and above its normal profit. This factor is called _____________.**
    A) Financial Health B) Cost estimate uncertainty
    C) Market opportunity D) Requirements volatility
18. **_________ is created that records the work to be done, who will do it, the development schedule, and the work products.**
    A) Decision B) Solution
    C) Project plan D) Strategy
19. **__________ is not a risk management activity.**
    A) Risk assessment B) Risk generation
    C) Risk control D) Risk monitoring
20. **Quality Management is essential for project ________.**
    A) failure B) degradation
    C) delay D) success

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I | | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|
| 1. | a) Why is software engineering important? Also, briefly explain with a neat diagram the various activities of requirements engineering process(RE). | 08 | L*1 | 1 | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | b) | Explain the incremental software process model with a neat diagram and discuss its merits and demerits. | 08 | L2 | 2 | 1 |
| 2. | a) | Define requirement. Also, discuss various types of non-functional requirements with suitable example for any software system of your choice. | 08 | L1 | 1 | 1 |
| | b) | Explain with a neat diagram spiral software process model. | 08 | L2 | 2 | 1 |
| 3. | a) | List the ACM/IEEE code of ethics that software engineers should follow to maintain professional ethical standards. Also, with example summarize an essential attributes of a good software. | 08 | L1 | 1 | 8 |
| | b) | Explain the waterfall software process model with a neat diagram and discuss its merits and demerits. | 08 | L2 | 2 | 1 |
| | | **Unit – II** | | | | |
| 4. | a) | Recall the principles of agile manifesto. | 04 | L1 | 4 | 1 |
| | b) | Explain test-driven development (TDD) with a suitable diagram. | 04 | L2 | 4 | 1 |
| | c) | Model the use-case diagram of a medical receptionist role in a mental health care patient monitoring system (MHC-PMS). Also, give the tabular description of transfer-data use case. | 08 | L3 | 3 | 2 |
| 5. | a) | What are the advantages of pair programming? | 04 | L1 | 4 | 1 |
| | b) | Explain briefly different stages in an object-oriented design process (OOD). | 06 | L2 | 3 | 1 |
| | c) | Model the sequence diagram for MHC – PMS transfer data use-case. | 06 | L3 | 3 | 2 |
| 6. | a) | Explain various forms of release testing with suitable example for each. | 08 | L2 | 4 | 1 |
| | b) | Model the state transition diagram for microwave oven operation. Also, provide tabular representation of various states of microwave oven. | 08 | L3 | 3 | 2 |
| | | **Unit – III** | | | | |
| 7. | a) | Show various factors affecting software pricing. | 04 | L1 | 5 | 1 |
| | b) | Interpret with suitable diagram about software review process. | 04 | L2 | 5 | 1 |
| | c) | Explain risk management process with suitable diagram. | 08 | L2 | 5 | 2 |
| 8. | a) | Recall various advantages of a cohesive group. | 04 | L1 | 5 | 1 |
| | b) | Illustrate the need of any four static software product metrics. | 04 | L2 | 5 | 1 |
| | c) | Explain project scheduling process with suitable diagram. | 08 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

*Off-Campus Centre of Nitte (Deemed to be University)*

**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

## CS3005-1– MICROPROCESSOR AND EMBEDDED SYSTEMS

(For CS, CB)

Duration: 3 Hours — Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **Which index register in the 8086 is primarily used for operations involving string processing?**
   A) BP, SP  B) DH, DL
   C) BH, BL  D) SI, DI
2. **8086 can access up to?**
   A) 512KB  B) 2Mb
   C) 1Mb  D) 256KB
3. The instruction, MOV AX, 0005H belongs to the address mode
   A) register  B) direct
   C) immediate  D) register indirect
4. **Match the following**
   **a) DB  1) used to direct the assembler to reserve only 10-bytes**
   **b) DT  2) used to direct the assembler to reserve only 4 words**
   **c) DW  3) used to direct the assembler to reserve byte or bytes**
   **d) DQ  4) used to direct the assembler to reserve words**
   A) a-3, b-2, c-4, d-1  B) a-2, b-3, c-1, d-4
   C) a-3, b-1, c-2, d-4  D) a-3, b-1, c-4, d-2
5. **Which group of instructions do not affect the flags?**
   A) Arithmetic operations  B) Logic operations
   C) Data transfer operations  D) Branch operations
6. **The instruction PUSHF is used to:**
   A) Push a register onto the stack  B) Push the flag register onto the stack
   C) Pop a register from the stack  D) Load a segment register
7. **If AX = 1001 0110 1010 1100B and CF = 1, what will be the result of RCL AX, 1?**
   A) 0010 1101 0101 1001B, CF = 1  B) 0010 1101 0101 1000B, CF = 1
   C) 0010 1101 0101 1001B, CF = 0  D) 1001 0110 1010 1101B, CF = 1
8. The instruction RET is used to:
   A) Return from subroutine  B) Jump to a memory address
   C) Load a segment register  D) Move to the next instruction
9. **Which function of INT 16H checks if a key is available?**
   A) AH = 00H  B) AH = 01H
   C) AH = 02H  D) AH = 10H
10. **What is the ASCII value of the hexadecimal digit 0x9?**
    A) 0x39  B) 0x49
    C) 0x09  D) 0x1001

11. Set time interrupt from DOS interrupt will store the hour information in which register
   A) CH B) CL
   C) DH D) DL
12. In a program, a Macro is being called 'n' times. Then how many times is the machine code generated for the same?
   A) 1 time B) 'n' times
   C) 'n-1' times D) 'n+1' times
13. Which register in ARM holds the flags indicating the status of the processor?
   A) General-purpose register B) Program Counter (PC)
   C) Stack Pointer (SP) D) Program Status Register (PSR)
14. How does the 7-segment display represent numbers?
   A) Using rows and columns of LEDs B) By turning on specific segments of LEDs
   C) Through alphanumeric LCD cells D) By activating relays
15. Which built-in Arduino function is used for serial communication?
   A) digitalWrite() B) Serial.begin()
   C) analogRead() D) pinMode()
16. What is the primary design philosophy of ARM embedded systems?
   A) Complex Instruction Set Computing (CISC) B) Reduced Instruction Set Computing (RISC)
   C) Very Long Instruction Word (VLIW) D) Superscalar Processing
17. Which pin is used by the 8086 to indicate that an external device is ready to exchange data?
   A) READY B) INTR
   C) ALE D) HLDA
18. Which bit in the MOV instruction encoding specifies whether the data transfer is of a byte or a word?
   A) D (direction) bit B) MOD field bit
   C) W (word) bit D) REG field bit
19. During an interrupt response, which register is the first to be pushed onto the stack by the 8086?
   A) CS (Code Segment) B) IP (Instruction Pointer)
   C) FLAGS D) DS (Data Segment)
20. During a read cycle, when are the address lines present on the multiplexed bus?
   A) After the data is transferred B) During the WR signal activation
   C) When the ALE signal is active D) When the HLDA signal is active

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

| | | Unit – I | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|---|
| 1. | a) | Identify the addressing modes of the instructions given below and justify the answer with clear explanation.<br>i) MOV WORD PTR[SI], 50H<br>ii) MOV DS:[1000H], 10H<br>iii) MOV AX, Num[BX+DI] | 6 | L*3 | 1 | 2 |
| | b) | Write an Assembly level Program to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm. | 6 | L3 | 2 | 2 |
| | c) | Explain the use of instruction queue in 8086 architecture. | 4 | L2 | 1 | 1 |
| 2. | a) | Describe the roles of various status flags and Control flags in Intel 8086 Microprocessor along with the flag register format. If the two numbers 1100H and 60C9H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer. | 6 | L3 | 1 | 1 |

| | | | Marks | BT*L* | CO* | PO* |
|---|---|---|---|---|---|---|
| | b) | How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example. | 6 | L1 | 2 | 1 |
| | c) | Write an Assembly level Program to display your name in front of the message "what is your name". | 4 | L3 | 2 | 2 |
| 3. | a) | Explain the working of following 8086 instructions along with example for each:<br>i) AAM ii) DAA | 6 | L2 | 2 | 1 |
| | b) | Explain the following assembler directives with example:<br>i. ASSUME<br>ii. EQU<br>iii. DD | 6 | L2 | 1 | 1 |
| | c) | Assume that SI is 3000H and the content of 2FFFH = 04H; content of 3000H = 4EH; Content of 3001H = 9FH. Which value is stored in AX register after execution of the statement MOV AX, [SI]. Justify your answer. | 4 | L3 | 2 | 2 |
| | | **Unit – II** | | | | |
| 4. | a) | Briefly explain the Embedded system software components. | 8 | L2 | 4 | 1 |
| | b) | Write an Embedded C program to control the speed of rotation of a stepper motor using potentiometer. | 4 | L3 | 4 | 2 |
| | c) | Explain the operation of the following DOS INT 21H functions:<br>i) Function 3AH<br>ii) Function 2CH | 4 | L2 | 3 | 1 |
| 5. | a) | With code snippet, explain Serial.begin(),Serial.available(),Serial.println() functions. | 6 | L2 | 4 | 1 |
| | b) | Differentiate between Macro and a Procedure. Write a procedure to read two digit BCD number from keyboard and store in BL register. | 6 | L3 | 4 | 1 |
| | c) | Compare and contrast RISC and CISC Architecture. | 4 | L2 | 3 | 1 |
| 6. | a) | With a basic layout diagram, explain the ARM program status register by highlighting the functionalities of different fields. | 8 | L2 | 4 | 1 |
| | b) | Write an Embedded C program to perform up-down counter on a 7-segment display using switches. | 4 | L3 | 4 | 2 |
| | c) | Write an Assembly Language Program to compute the factorial of a positive integer 'n' using a recursive procedure. | 4 | L3 | 3 | 2 |
| | | **Unit – III** | | | | |
| 7. | a) | Categorize the following signals into minimum mode signals and maximum mode signals. Write their roles.<br>(i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0' (iii) ALE (iv) DEN (v)HLDA | 8 | L1 | 5 | 1 |
| | b) | Explain the coding template for the MOV instruction of 8086 Intel microprocessor and Generate machine code for the following instructions of Intel 8086 microprocessor<br>MOV AL, [1000] [BX] | 8 | L3 | 5 | 2 |
| 8. | a) | List the 8086 signals having common functions in both minimum and maximum mode and explain any four signals in detail. | 10 | L1 | 5 | 1 |
| | b) | What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. | 6 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

## Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

## CS3005-1– MICROPROCESSOR AND EMBEDDED SYSTEMS

(CS)

ation: 3 Hours

te:

Max. Marks: 100

t – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each estion carries equal marks.

t – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from it – I & **Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS

**20 Marks**

The 1 MB byte of memory can be divided into segment

A) 1 Kbyte  B) 24 Kbyte  
C) 32 Kbyte  D) 64 Kbyte

The intel 8086 microprocessor is a processor

A) 8 bit  B) 16 bit  
C) 32 bit  D) 4 bit

The JS is called as

A) jump signed bit  B) jump single bit  
C) jump simple bit  D) jump signal it

The expansion of DAA is

A) decimal adjust after addition  B) decimal adjust before addition  
C) decimal adjust accumulator  D) decimal adjust auxiliary

The instruction that is invalid among the following is

A) MOV AX, BX  B) MOV AX, [BX]  
C) MOV 55H, BL  D) MOV AL, 55H

The directive used to inform the assembler, the names of the logical segments to be assumed for different segments used in the program is

A) DB  B) SEGMENT  
C) SHORT  D) ASSUME

In the RCL instruction, the contents of the destination operand undergo function as

A) carry flag is pushed into LSB & MSB is pushed into the carry flag  
B) carry flag is pushed into MSB & LSB is pushed into the carry flag  
C) auxiliary flag is pushed into LSB & MSB is pushed into the carry flag  
D) auxiliary flag is pushed into MSB & LSB is pushed into the carry flag

NMI stands for

A) Non Movable Interrupt  B) Non Multiple Input  
C) Non Maskable Interrupt  D) Non Maskable Input

An interrupt breaks the execution of instructions and diverts its execution to

A) Interrupt service routine  B) Counter word register  
C) Execution unit  D) control unit

When the CPU executes IRET

A) the control transfers from ISR to main program  B) contents of IP and CS are retrieved  
C) clears the trap flag  D) clears the interrupt flag

The address bits are sent out on lines through

A) $A_{16}$- $A_{19}$  B) $AD_0 - AD_{16}$  
C) C0-C17  D) $AD_0$- $AD_{15}$ and $A_{16} - A_{19}$

12. What is DEN?
A) direct enable
B) data entered
C) data enable
D) data encoding

13. Which group of instructions do not affect the flags?
A) Arithmetic operations
B) Logic operations
C) Data transfer operations
D) Branch operations

14. Arduino IDE consists of 2 functions. What are they?
A) build() and setup()
B) build() and loop()
C) setup() and build()
D) setup() and loop()

15. What is the output of "pin1" if "pin2" is sent "1011" where 1 is 5V and 0 is 0V?

```
int pin1 = 12;
int pin2 = 11;
void setup() {
   pinMode(pin1, OUTPUT);
   pinMode(pin2, INPUT);
   Serial.begin(9600);
}
void loop() {
   if(digitalRead(pin2)==1) {
      digitalWrite(pin1,LOW);
   }
   else if(digitalRead(pin2)==0) {
      digitalWrite(pin1,HIGH);
   }
}
```

A) 1110
B) 0100
C) 1111
D) 1011

16. What is the purpose of the following Arduino code?

```
void setup() {
   Seria.begin(9600);
}
void setup() {
   Serial.write(40);
}
```

A) Send a signal to pin 40 on the Arduino board
B) Send a octal number of 40 through the Serial pins
C) Send a byte with value 40 through the Serial pins
D) Send a hexadecimal number of 40 through the Serial pins

17. How many times does the setup() function run on every startup of the Arduino System?
A) 1
B) 2
C) 3
D) 4

18. Can the loop() function be used to call another function that is custom defined by the programmer?
A) Yes, it can call but only functions with return values
B) No, it cannot call
C) Yes, it can call but only functions with no return values
D) Yes, it can call

19. What are the two modes that the pinMode() method sets for a particular pin?
A) INPUT and OUTPUT
B) TX and RX
C) DIGITAL and ANALOG
D) READ and WRITE

20. What type of signal does the analogWrite() function output?
A) Pulse Code Modulated Signal
B) Pulse Width Modulated Signal
C) Pulse Amplitude Modulated Signal
D) Frequency Modulated Signal

Summer Semester July 2024

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

### Unit – I

| Q. No. | | Question | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|---|
| 1. | a) | Write an ALP to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm. | 6 | L*3 | 1 | 1 |
| | b) | What is the difference between the following instructions?<br>(i) JL UP and JB UP (ii) XOR AX, AX and MOV AX, 0 | 4 | L2 | 1 | 1 |
| | c) | State the various assembler directives used and their meaning in the assembly language segment given below<br>OUR_DATA SEGMENT<br>NUM1 DW 1234H<br>NUM2 DB 12H,34H,3DUP(0)<br>OUR_DATA ENDS | 6 | L2 | 1 | 1 |
| 2. | a) | How various Shift Instructions work? Differentiate between arithmetic and logical shift instructions. Explain with examples. | 6 | L2 | 1 | 1 |
| | b) | Write an ALP to exchange two register contents AL and BL without using XCHG instruction and without using temporary register/memory location. | 4 | L3 | 1 | 1 |
| | c) | Explain the flag register along with register format with respect to Intel 8086. | 6 | L1 | 1 | 1 |
| 3. | a) | Identify the addressing modes of following instructions and justify the answer<br>(i) MOV WORD PTR [SI], 20H<br>(ii) MOV ES:[1000H], 10H<br>(iii) MOV CX, NUM[BX + DI] | 6 | L2 | 1 | 1 |
| | b) | Explain the use of instruction queue in 8086 architecture. | 4 | L1 | 1 | 1 |
| | c) | Write an ALP to reverse a given string and check whether it is a palindrome or not. | 6 | L3 | 1 | 1 |

### Unit – II

| Q. No. | | Question | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|---|
| 4. | a) | Write a Program to perform following operations:<br>i) Create a file<br>ii) Write "GOOD MORNING" in created file | 4 | L3 | 2 | 1,2 |
| | b) | Develop an embedded C program to perform up-down counter on a 7-segment display using switches. | 6 | L3 | 4 | 1 |
| | c) | Describe the program status register of ARM processor with neat diagram. | 6 | L2 | 3 | 1,2 |
| 5. | a) | Write an ALP to do the following BIOS and DOS operations:<br>Read a character from the KeyBoard. Add 4 lines and 4 columns to the current cursor position. Clear the entire screen and bring the cursor to newly calculated position and display the ASCII value of pressed character. | 4 | L3 | 2 | 1,2 |
| | b) | With code snippet, explain digitalWrite(), and analogRead() functions. | 4 | L1 | 4 | 1 |
| | c) | Explain the ARM core model and its functional units with a neat diagram. | 8 | L2 | 3 | 1,2 |
| 6. | a) | Write an ALP to convert two ASCII values to the equivalent hexadecimal. | 4 | L3 | 2 | 1,2 |
| | b) | What is recursion? Write an ALP to find factorial of a number stored in data segment using recursion. | 4 | L2 | 2 | 1,2 |
| | c) | Write an Embedded C program to read the input from 4x4 keypad and simulate operations as a calculator. | 8 | L3 | 4 | 1 |

## Unit – III

| | | | | Marks | BT | CO | PO |
|---|---|---|---|---|---|---|---|
| 7. | a) | Give the coding template for the MOV instruction of the 8086 Intel microprocessor. Generate machine code for the following instructions of the Intel 8086 microprocessor.<br>MOV CL, [1101] | | 6 | L3 | 5 | 2 |
| | b) | Explain the Type 3 and Type 4 interrupts in 8086. | | 4 | L1 | 5 | 2 |
| | c) | Differentiate between Minimum mode and maximum mode operation of 8086 microprocessor. | | 6 | L2 | 5 | 2 |
| 8. | a) | Generate machine code for the following instructions of Intel 8086 microprocessor.<br>MOV CX, 4012H[DI+BX] | | 2 | L3 | 5 | 2 |
| | b) | What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. | | 4 | L1 | 5 | 2 |
| | c) | List the signals used in both minimum and maximum mode operation and explain any four signals in detail. | | 10 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

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**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

## MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS

(For CS, IS, AM, CC,CB,RI)

Duration: 3 Hours Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS **20 Marks**

1. **System of equations AX=B has infinite number of solutions if**
   A) Rank[A] = Rank[A|B] = number of unknowns
   B) Rank[A] = Rank[A|B] < number of unknowns
   C) Rank[A] ≠ Rank[A|B]
   D) Rank[A] = Rank[A|B] > number of unknowns

2. **If** $A=\begin{pmatrix}-1 & 1\\ 2 & 4\end{pmatrix}$ **then** $tr(AA^T)$ **is**
   A) 21
   B) 22
   C) 3
   D) 26

3. **The eigen values of** $A=\begin{bmatrix}3 & 2\\ 3 & 8\end{bmatrix}$ **are**
   A) 7,4
   B) 9,2
   C) 8,3
   D) 6,5

4. **If A is** $3\times 4$ **matrix the maximum possible value of the rank A is**
   A) 4
   B) 3
   C) 2
   D) 0

5. **The characteristic equation of the matrix** $A=\begin{bmatrix}2 & 0\\ 4 & 7\end{bmatrix}$ **is:**
   A) $\lambda^2-9\lambda+14=0$
   B) $\lambda^2-14\lambda+9=0$
   C) $\lambda^2-9\lambda-14=0$
   D) $\lambda^2-14\lambda-9=0$

6. **Number of pivot columns in the matrix** $\begin{bmatrix}0 & 0 & 1 & 0\\ 2 & 0 & 1 & 0\\ 1 & 0 & 0 & 0\end{bmatrix}$ **is:**
   A) 1
   B) 2
   C) 3
   D) 4

7. **Which of the following method is not used to find the solution of the system of equations:**
   A) Gauss elimination method
   B) Gauss Seidel method
   C) Rayleigh's power method
   D) LU decomposition method

8. **If the LU factorization of a matrix A is A=LU where A=**$\begin{bmatrix}2 & 5\\ -3 & -4\end{bmatrix}$ **then U is**
   A) $\begin{bmatrix}2 & 5\\ 0 & 7\end{bmatrix}$
   B) $\begin{bmatrix}1 & 4\\ -5 & 1\end{bmatrix}$
   C) $\begin{bmatrix}2 & 5\\ 1 & 7\end{bmatrix}$
   D) $\begin{bmatrix}1 & 4\\ -7/2 & 1\end{bmatrix}$

9. **If** $\{u_1,u_2,\ldots\ldots\ldots,u_n\}$**is linearly independent set, then** $c_1u_1+c_2u_2+\ldots\ldots+c_nu_n=0$ **has**
   A) infinitely many solutions
   B) unique nonzero solution
   C) unique zero solution
   D) both B and C are possible

10. The row space of an $m \times n$ matrix A is a subspace of
   A) $R^n$ B) A
   C) $R^m$ D) R

11. If $S = \{v_1, v_2, .., v_n\}$ is a set of vectors in a finite dimension vector space $V$, then $S$ is called a basis for $V$ if
   A) $S$ spans $V$ B) $S$ is linearly independent
   C) either (A) or (B) D) both (A) and (B)

12. Let $T: R^2 \to R^2$ be the linear transformation defined by $T(x, y) = (0, y)$. Then matrix of linear transformation with respect to standard basis is
   A) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ B) $\begin{bmatrix} 0 & -1 \\ 0 & -1 \end{bmatrix}$
   C) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ D) $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$

13. The set of all solution to the homogeneous equation system $AX = 0$, where A be an $m \times n$ matrix is
   A) Null space of A B) Column space of A
   C) Row space of A D) Null space of A and column space of A

14. If a basis of a vector space has 3 elements, then dimension of the vector space is_______
   A) less than 4 B) greater than 3
   C) 3 D) 0

15. If $u = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}, v = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$ then inner product of $u$ and $v$ is
   A) -5 B) -11
   C) 19 D) 11

16. Let $T: R^3 \to R^3$ be the linear transformation with nullity of T as 1 then rank of T is
   A) 1 B) 2
   C) 3 D) 4

17. In PCA, what is the relationship between the first principal component and the second principal component
   A) They are orthogonal to each other B) They are positively correlated
   C) They are negatively correlated D) There is no defined relationship between them

18. The matrix of the quadratic form $4x_1x_2 + 6x_1x_3 - 8x_2x_3$
   A) $\begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & -4 \\ 2 & -4 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 0 & 1 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$
   C) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -4 \\ 3 & -4 & 1 \end{bmatrix}$ D) $\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$

19. Let $B$ be the mean deviation form of N observation vector then covariance matrix is
   A) $S = \frac{1}{N-1} B^T$ B) $S = \frac{1}{N-1} BB^T$
   C) $S = \frac{1}{N+1} BB^T$ D) $S = \frac{1}{N} BB^T$

20. Singular values of $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ are
   A) 1,-3 B) 9,1
   C) $\sqrt{3}$ ,1 D) 3,1

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

| | | | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|---|
| | | **Unit – I** | | | | |
| 1. | a) | Diagonalize the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$.Hence find $A^6$. | 6 | L*3 | 2 | 2 |
| | b) | Use Gauss-Seidel iteration method to solve the following system of linear equations $20x + y - 2z = 17, 3x + 20y - z = -18$, 2x-3y+20z=25, start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations. | 6 | L3 | 1 | 2 |
| | c) | Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$ | 4 | L3 | 1 | 2 |
| 2. | a) | Find the value of **a and b** so that the equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x$ +3y+az=b has i) no solution ii) a unique solution iii) more than one solution. | 6 | L3 | 1 | 2 |
| | b) | Solve the system of equations $x + y + z = 1$, $4x + 3y - z = 6,\ 3x + 5y + 3z = 4$ by LU factorization method. | 6 | L3 | 1 | 2 |
| | c) | Using the power method, find the dominant eigen value and corresponding eigen vector of the matrix $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ starting with the initial value $[1 \quad 0 \quad 0]^T$. Carry out four iterations. | 4 | L3 | 2 | 2 |
| 3. | a) | Find all the eigen values of $A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$.Find an eigen vector corresponding to the largest eigen value. | 6 | L3 | 1 | 2 |
| | b) | Reduce the given matrix A to echelon form and find its rank A= $\begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$ | 6 | L3 | 1 | 2 |
| | c) | Define Trace of a matrix. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$ then verify that i) $tr(A + B) = tr(A) + tr(B)$ ii)$tr(AB) = tr(BA)$ | 4 | L3 | 2 | 2 |
| | | **Unit – II** | | | | |
| 4. | a) | Let $T: R^3 \rightarrow R^2$ be a linear map defined by $T(x, y, z) = (y - x, y - z)$. Find the null space and range space of T. Also find dimension of null space and range space of T. | 6 | L3 | 4 | 2 |
| | b) | Find a basis for the row space of $\begin{bmatrix} 1 & -3 & 4 & -1 & 9 \\ -2 & 6 & -6 & -1 & -10 \\ -3 & 9 & -6 & -6 & -3 \\ 3 & -9 & 4 & 9 & 0 \end{bmatrix}$ | 6 | L3 | 3 | 2 |
| | c) | Check whether the set of vectors $\{(1,2,3), (4,5,6), (2,1,0)\}$ is linearly dependent or linearly independent. | 4 | L3 | 3 | 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5. | a) | Prove that a nonempty subset W of a vector space V over the field F is a subspace if and only if for any $\alpha, \beta \in W$ and $c \in F, c\alpha + \beta \in W$. | 6 | L1 | 3 | 1 |
| | b) | Let $W = span\{x_1, x_2, x_3\}$ where $x_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$, $x_2 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix}$, $x_3 = \begin{bmatrix}0\\0\\1\\1\end{bmatrix}$ Construct an orthogonal basis for W using Gram -Schmidt process. | 6 | L3 | 4 | 2 |
| | c) | Find the co-ordinate vector of $(2,7,-4) in\ R^3$ with respect to the basis $\{(1,2,0),(1,3,2),(0,1,3)\}$. | 4 | L3 | 3 | 2 |
| 6. | a) | Determine whether (1,1,1,1), (1,2,3,2), (2,5,6,4), (2,6,8,5) form a basis of $R^4$. If not find the dimension of the subspace they span. | 6 | L3 | 3 | 2 |
| | b) | Prove that a set of vector $\{v_1, v_2, \dots\dots\dots, v_n\}$ is linearly dependent if and only if at least one of them is linear combination of the others. | 6 | L1 | 3 | 1 |
| | c) | Find the matrix of linear transformation $T: V_2(R) \to V_3(R)$ defined by $T(x,y) = (x+y, x, 3x-y)$ with respect to the bases $B_1 = \{(1,1),(3,1)\}$ and $B_2 = \{(1,1,1),(1,1,0),(1,0,0)\}$. | 4 | L3 | 4 | 2 |

**Unit – III**

| | | | | | | |
|---|---|---|---|---|---|---|
| 7. | a) | Find QR factorization of matrix $A = \begin{bmatrix}1 & 3 & 5\\1 & 1 & 0\\1 & 1 & 2\\1 & 3 & 3\end{bmatrix}$ | 8 | L3 | 5 | 2 |
| | b) | Find a least square solution of AX=B for $A = \begin{bmatrix}1 & 1 & 0\\1 & 1 & 0\\1 & 0 & 1\\1 & 0 & 1\end{bmatrix}$, $B = \begin{bmatrix}1\\3\\8\\2\end{bmatrix}$ | 8 | L3 | 5 | 2 |
| 8. | a) | Find a singular value decomposition of matrix $A = \begin{bmatrix}1 & -1\\-2 & 2\\2 & -2\end{bmatrix}$ | 10 | L3 | 5 | 2 |
| | b) | Convert the matrix $\begin{bmatrix}1 & 5 & 2 & 6 & 7 & 3\\3 & 11 & 6 & 8 & 15 & 11\end{bmatrix}$ of observations to mean deviation form and construct the sample covariance matrix. | 6 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

*Off-Campus Centre of Nitte (Deemed to be University)*

**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

**CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS**

(For CS, IS , AM, CC, AD,CB)

Duration: 3 Hours | Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **How many sub arrays does the quick sort algorithm divide the entire array into?**
   A) One B) Two
   C) Three D) Four
2. **Binary Search works only on which type of arrays?**
   A) Unsorted arrays B) Sorted arrays
   C) Arrays with distinct elements D) Arrays of integers
3. **Linear search algorithm, also known as____**
   A) Binary search B) Simple search
   C) Sequential search D) Difficult search
4. **In asymptotic analysis, what does it mean if a function is said to be O(n )?**
   A) The time complexity increases exponentially as n increases
   B) The algorithm's runtime grows linearly with the input size
   C) The runtime of the algorithm is independent of the input size
   D) The algorithm's runtime grows logarithmically as n increases
5. **What will be the output of the following pseudocode if the driver is fun(5)? function fun(n) if n == 1 return 1 else return n + fun(n-1)**
   A) 120 B) 24
   C) 6 D) 15
6. **Pseudocode is written using**
   A) Combination of Native & programming language
   B) High level Programming language only
   C) Natural language only
   D) Low level programming language
7. **Which of the following is NOT a characteristic of a good algorithm?**
   A) Clarity and simplicity B) Efficiency in terms of time and space
   C) Correctness D) The use of complex data structures
8. **What is the primary characteristic of the Divide and Conquer approach?**
   A) Solving problems by iteration
   B) Solving problems by dividing them into overlapping sub problems
   C) Dividing a problem into smaller independent subproblems, solving them recursively, and combining their results
   D) Dividing a problem into smaller problems and linearly solving it.
9. **Which technique is used to solve the 0/1 Knapsack problem efficiently?**
   A) Greedy algorithm B) Divide and conquer
   C) Dynamic programming D) Depth-first search

**10. What is the primary purpose of the Floyd-Warshall algorithm?**

A) Finding the shortest path between two specific nodes
B) Finding the shortest paths between all pairs of nodes in a graph
C) Finding the minimum spanning tree of a graph
D) Finding the longest path in a weighted graph

**11. In Horspool's algorithm, which table is used to store the shift values for the pattern matching process?**

A) Shift Table
B) Hash Table
C) Match Table
D) Character Table

**12. What makes Counting Sort a distribution-based algorithm?**

A) It uses element frequencies to determine sorted positions.
B) It divides the array into smaller subarrays.
C) It recursively splits and merges data.
D) It compares each pair of elements to sort the array.

**13. You are given an array: [3, 9, 2, 1, 4, 8]. After building a max-heap from this array, what will be the root element?**

A) 1
B) 9
C) 2
D) 8

**14. What operation is performed on an AVL tree when the balance factor of a node becomes greater than 1 or less than -1?**

A) Rotation
B) Deletion
C) Insertion
D) Search

**15. Binary Search is a classic example of which variation of Decrease and Conquer?**

A) Decrease by a constant
B) Decrease by a constant factor
C) Decrease by one
D) Divide and conquer

**16. In a max-heap, which of the following is true?**

A) The parent node is smaller than its child nodes
B) The parent node is greater than or equal to its child nodes
C) The tree is a binary search tree
D) The elements are stored in ascending order

**17. In the Job Assignment Problem, what does the objective function typically aim to minimize?**

A) The time to complete all tasks
B) The number of tasks assigned
C) The total cost of assigning tasks to workers
D) The total time worked by all workers

**18. In the subset-sum problem, when do we backtrack?**

A) When the sum of the selected elements exceeds the target sum
B) When the sum of the selected elements is equal to the target sum
C) When all elements have been considered
D) When a subset has been found

**19. Dijkstra's Algorithm is the prime example for ___________**

A) Greedy algorithm
B) Branch and bound
C) Back tracking
D) Dynamic programming

**20. Which of the following algorithms is the best approach for solving Huffman codes?**

A) exhaustive search
B) greedy algorithm
C) brute force algorithm
D) divide and conquer algorithm

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

| | Unit – I | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|
| 1. a) | Explain the Performance Analysis of an algorithm. | 8 | L*2 | 1 | 1 |
| b) | Write the pseudocode to sort the elements in an array using selection sort and analyze its efficiency. Apply selection sort to sort the following elements in an array. 5, 4, 2, 1, 6, 3. | 8 | L3 | 2 | 2 |
| 2. a) | Write the general plan for analyzing the efficiency of non recursive algorithm. Write the pseudocode to find the maximal element in an array and analyze its efficiency. | 8 | L3 | 1 | 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | b) | Write Divide And Conquer recursive Merge sort algorithm and derive the time complexity of this algorithm. | 8 | L3 | 2 | 2 |
| 3. | a) | Write the algorithm for brute force string matching. Show the comparisons the Brute force string matching makes for the patternP=acbcac in the text T=acbcabccababcaacbcac. | 8 | L3 | 1 | 2 |
| | b) | Illustrate Quick sort algorithm. Sort the following list using Merge sort.38, 27, 43, 3, 9, 82, 10. | 8 | L3 | 2 | 2 |

**Unit – II**

| | | | | | | |
|---|---|---|---|---|---|---|
| 4. | a) | Write an insertion sorting algorithm using decrease and conquer technique. Apply the algorithm on the following numbers and trace the steps - 23,1,10,5,2 | 8 | L3 | 3 | 2 |
| | b) | Construct AVL tree for the following input by showing each step of the insertion process and type of rotation used for transformation. 63, 9, 19, 27, 18, 108, 99, 81 | 8 | L3 | 4 | 2 |
| 5. | a) | Identify the pattern "BARBER" in the given text "JIM_SAW_ME_IN_A_BARBERSHOP" using Horspool's string-matching algorithm. Show the construction of shift table for the same. | 8 | L3 | 3 | 2 |
| | b) | Write an algorithm for sorting list of integers by comparison method that uses input enhancement design technique. Apply the algorithm on the following numbers and trace the steps 25,45,10,20,50,15. | 8 | L3 | 4 | 2 |
| 6. | a) | Define topological sorting. Apply DFS based method to topologically sort the vertices of the following graph. | | | | |

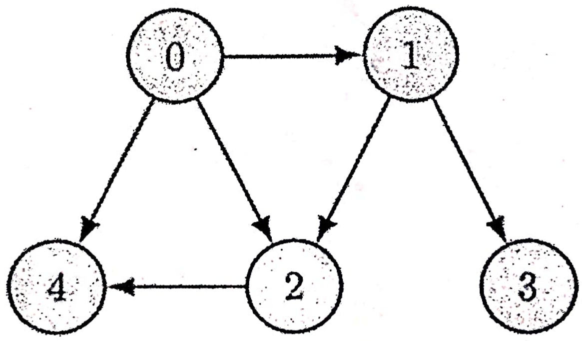


Fig. 6a — 8 L3 3 2

b) Write and Apply Floyd's all pairs shortest path algorithm for the graph shown in the following figure.

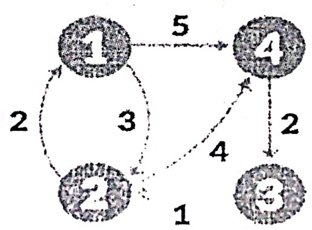


Fig. 6b — 8 L3 4 2

**Unit – III**

| | | | | | | |
|---|---|---|---|---|---|---|
| 7. | a) | Solve the given instance of sum of subset problem S={5,10,12,13,15,18} and d=30. Construct a state space tree. | 8 | L3 | 5 | 2 |
| | b) | Define minimum cost spanning tree. Write Prim's algorithm to find minimum cost spanning tree. | 8 | L2 | 5 | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 8. | a) | Explain the following: (i) Class P (ii) Class NP (iii) NP Complete Problem (iv) NP Hard Problem. | 8 | L2 | 5 | 1 |
| | b) | Solve the job assignment problem using Branch and Bound technique for the following matrix. | 8 | L3 | 5 | 2 |

| | Job 1 | Job 2 | Job 3 | Job 4 |
|---|---|---|---|---|
| A | 9 | 2 | 7 | 8 |
| B | 6 | 4 | 3 | 7 |
| C | 5 | 8 | 1 | 8 |
| D | 7 | 6 | 9 | 4 |

Fig. 8b

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

***Off-Campus Centre of Nitte (Deemed to be University)***

**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

**CS2102-1 – DATABASE MANAGEMENT SYSTEMS**

(For CS, IS, AM, CC, AD)

Duration: 3 Hours | Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **Which of the following best describes a database?**
   A) A collection of unrelated data
   B) A structured collection of related data stored for retrieval
   C) A set of tables without any relationships
   D) A file system
2. **In ER-to-Relational mapping, what does each entity type translate into?**
   A) A relationship
   B) A foreign key
   C) A table
   D) A schema
3. **Which type of attribute cannot be divided into smaller parts?**
   A) Composite attribute
   B) Derived attribute
   C) Multivalued attribute
   D) Simple attribute
4. **Which of the following attributes is a derived attribute?**
   A) Date of birth
   B) Age
   C) Name
   D) Address
5. **In an ER model, what does a diamond symbol represent?**
   A) Attribute
   B) Entity
   C) Relationship
   D) Primary Key
6. **A relationship where an entity is related to itself is called:**
   A) Binary Relationship
   B) Recursive Relationship
   C) Ternary Relationship
   D) Self-Relationship
7. **For each attribute of a relation, there is a set of permitted values, called the ________ of that attribute.**
   A) Domain
   B) Relation
   C) Set
   D) Schema
8. **What SQL statement would you use to create SQL database named Test?**
   A) CREATE DBT Test;
   B) CREATE Test
   C) CREATE DATABASE Test;
   D) MAKE DATABASE Test;
9. **How can you change "Thomas" into "Michel" in the "LastName" column in the Users table?**
   A) UPDATE User SET LastName = 'Thomas' INTO LastName = 'Michel'
   B) MODIFY Users SET LastName = 'Michel' WHERE LastName = 'Thomas'
   C) MODIFY Users SET LastName = 'Thomas' INTO LastName = 'Michel'
   D) UPDATE Users SET LastName = 'Michel' WHERE LastName = 'Thomas'
10. **Which command is used to change the definition of a table in SQL?**
    A) CREATE
    B) UPDATE
    C) ALTER
    D) SELECT

11. **What is the result of the query SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE;?**
    A) Total number of employees
    B) Total number of unique salaries
    C) Total number of salaries greater than a specified value
    D) Average salary of employees

12. **EXISTS operator is used with WHICH of the following clause(s)?**
    A) Select
    B) In
    C) Where
    D) Group by

13. **What is the primary use of a SQL view?**
    A) To store permanent data in the database
    B) To create a virtual table from a SQL query
    C) To create relationships between tables
    D) To perform database migrations

14. **What happens when DROP TABLE CASCADE is executed?**
    A) Only the table structure is removed
    B) The table and its dependent objects are removed
    C) The table data is cleared, but the structure is preserved
    D) The table is locked for further operations

15. **Which normal form ensures that no non-prime attribute is dependent on a part of a candidate key?**
    A) First Normal Form (1NF)
    B) Second Normal Form (2NF)
    C) Third Normal Form (3NF)
    D) Boyce-Codd Normal Form (BCNF)

16. **What is the time complexity of searching in a B+ Tree?**
    A) O(log n)
    B) O(n)
    C) O(1)
    D) O(n log n)

17. **Which indexing method is best for dynamic datasets?**
    A) Hash Indexing
    B) Bitmap Indexing
    C) B+ Tree Indexing
    D) Sorted File

18. **To generate the output of a query, a ___ is responsible**
    A) Query execution engine
    B) Query execution motor
    C) Query execution train
    D) None

19. **The "all-or-none" property is commonly referred to as ________**
    A) Isolation
    B) Durability
    C) Atomicity
    D) None of the mentioned

20. **A transaction that has not been completed successfully is called as _______**
    A) Compensating transaction
    B) Aborted transaction
    C) Active transaction
    D) Partially committed transaction

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I | | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|
| 1. | a) What are the Characteristics of Database approach? Explain. | 8 | L*2 | 1 | 1 |
| | b) With a neat diagram, explain the three Schema architecture. | 8 | L2 | 1 | 1 |
| 2. | a) Explain the following types of attributes with examples:<br>(i) Composite<br>(ii) Multivalued<br>(iii) Atomic | 6 | L2 | 1 | 1 |
| | b) Explain the Cardinality Ratio for Binary relationships with examples. | 10 | L2 | 1 | 1 |
| 3. | a) Consider the following Tables: | | | | |

STUDENT(SName, SUSN, SPhoneNo, SAddress, SAge, SCGPA)
COURSE (Ccode, Ctitle, SUSN,Sname,Cstaff)

| | | | | | | |
|---|---|---|---|---|---|---|
| | | (i) What is the Degree of the relation STUDENT?<br>(ii) What is the Degree of the relation COURSE?<br>(iii) Construct a Relational algebra to display all details of students with SCGPA>8.0.<br>(iv) Construct a Relational algebra to display names of all students who have taken up 'DBMS' course.<br>(v) Construct a relational algebra to display the names of all courses taken up by students with SCGPA>9.0. | 10 | L3 | 2 | 1 |
| | b) | Explain the steps in ER to Relational Mapping Algorithm. | 6 | L2 | 2 | 1 |

**Unit – II**

| | | | | | | |
|---|---|---|---|---|---|---|
| 4. | a) | Explain how you can Specify the Not Null, DEFAULT and CHECK constraints in SQL. | 6 | L2 | 2 | 1 |
| | b) | Explain the CASCADE and RESTRICT options of DROP TABLE command. | 6 | L2 | 2 | 1 |
| | c) | What are the uses of ALTER TABLE command? | 4 | L1 | 2 | 1 |
| 5. | a) | What are the Informal Design Guidelines for relational schema? Mention all and explain any 2. | 8 | L1 | 2 | 1 |
| | b) | What is a Functional Dependency? Explain with an example. | 8 | L2 | 2 | 1 |
| 6. | a) | Explain the Standard Inference rules for Functional Dependencies. | 6 | L1 | 2 | 1 |
| | b) | Explain the algorithm for determining X+, the Closure of X under F. | 10 | L2 | 2 | 1 |

**Unit – III**

| | | | | | | |
|---|---|---|---|---|---|---|
| 7. | a) | What is a B+ Tree? Explain structure & characteristics of a B+ Tree with a neat diagram. | 10 | L1 | 3 | 1 |
| | b) | Briefly explain the ACID properties. | 6 | L2 | 3 | |
| 8. | a) | What is a Query Optimizer ? Explain its working with a neat diagram. | 10 | L1 | 3 | |
| | b) | What are the 2 rules of Strict 2PL ? Briefly explain. | 6 | L2 | 3 | |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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# NMAM INSTITUTE OF TECHNOLOGY, NITTE

*Off-Campus Centre of Nitte (Deemed to be University)*

**Fourth Semester B.Tech (CBCS) Degree Examinations**

Summer Semester July 2024

**CS2102-1– DATABASE MANAGEMENT SYSTEMS**

**(CS, IS, AI&ML, CCE, AI&DS)**

Duration: 3 Hours | Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

## PART - A: MULTIPLE CHOICE QUESTIONS — 20 Marks

1. **What is the full form of DBMS?**
   A) Data of Binary Management System
   B) Database Management System
   C) Database Management Service
   D) Data Backup Management System
2. **What is information about data called?**
   A) Hyper data
   B) Tera data
   C) Meta data
   D) Relations
3. **_____________ is a set of one or more attributes taken collectively to uniquely identify a record.**
   A) Primary Key
   B) Super key
   C) Foreign key
   D) Candidate key
4. **Which command is used to remove a relation from an SQL?**
   A) Drop table
   B) Delete
   C) Purge
   D) Remove
5. **_______________ operations do not preserve non-matched tuples.**
   A) Left outer join
   B) Inner join
   C) Natural join
   D) Right outer join
6. **Which forms have a relation that contains information about a single entity?**
   A) 4NF
   B) 5NF
   C) 2NF
   D) 3NF
7. **After groups have been established, SQL applies predicates in the __________ clause, allowing aggregate functions to be used.**
   A) Where
   B) Having
   C) Group by
   D) With
8. **Which of the following resembles Create view**
   A) Create table . . . as
   B) Create view as
   C) Create table . . .like
   D) With data
9. **When the "ROLLUP" operator for expression or columns within a "GROUP BY" clause is used?**
   A) Find the groups that make up the subtotal in a row
   B) Create group-wise grand totals for the groups indicated in a GROUP BY clause
   C) Group expressions or columns specified in a GROUP BY clause in one direction, from right to left, for computing the subtotals
   D) To produce a cross-tabular report for computing subtotals by grouping phrases or columns given within a GROUP BY clause in all available directions

10. Consider the relation schema: Singer(singerName, songName). What is the highest normal form satisfied by the "Singer" relation schema?
    A) 1NF B) 2NF
    C) BCNF D) 3NF
11. Which of the following is used to denote the selection operation in relational algebra?
    A) Pi (Greek) B) Lambda (Greek)
    C) Omega (Greek) D) Sigma (Greek)
12. Which is a join condition contains an equality operator:
    A) Equijoins B) Cartesian
    C) Natural D) Left
13. Which of the following makes the transaction permanent in the database?
    A) View B) Commit
    C) Rollback D) Flashback
14. A relation in which every non-key attribute is fully functionally dependent on the primary key and which has no transitive dependencies is said to be in
    A) BCNF B) 2NF
    C) 3NF D) 4NF
15. In order to maintain the consistency during transactions, database provides
    A) Commit B) Atomic
    C) Flashback D) Retain
16. _________ will undo all statements up to commit?
    A) Transaction B) Flashback
    C) Rollback D) Abort
17. Materialized views make sure that
    A) View definition is kept stable B) View definition is kept up-to-date
    C) View definition is verified for error D) View is deleted after specified time
18. The ______________ is essentially used to search for patterns in target string.
    A) Like Predicate B) Null Predicate
    C) In Predicate D) Out Predicate
19. Which of the following is the syntax for views where v is view name?
    A) Create view v as "query name"; B) Create "query expression" as view;
    C) Create view v as "query expression"; D) Create view "query expression";
20. The number of attributes in relation is called as its
    A) Cardinality B) Degree
    C) Tuples D) Entity

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

### Unit – I

| | | | Marks | BT* | CO* | PO* |
|---|---|---|---|---|---|---|
| 1. | a) | Briefly explain the various characteristics of the database approach. | 8 | L*1 | 1 | 1 |
| | b) | Explain the following Relational Algebra operations with example for each:<br>i. Division<br>ii. Union<br>iii. Cartesian product<br>iv. COUNT aggregate function | 8 | L2 | 2 | 1,2 |
| 2. | a) | Design an ER diagram and schema for the flight management system. Specify any assumptions made. | 8 | L3 | 1 | 1 |
| | b) | Explain grouping concepts in relational algebra. Give one example. | 4 | L2 | 2 | 1,2 |
| | c) | Explain following aggregate functions used in relational algebra. Give example for each.<br>i. AVERAGE<br>ii. COUNT | 4 | L2 | 2 | 1,2 |

3. a) With a neat block diagram explain the three schema architecture for databases. 8 L1 1 1

b) Consider the Sailors-Boats-Reserves DB described Below
SAILOR (sid, sname, rating, age)
BOAT (bid, bname, color)
RESERVES(sid, bid, date)
Write each of the following queries in Relational Algebra.
   i. Find all sailor id's of sailors who have a rating of at least 8 or reserved boat 103.
   ii. Find the names of sailors who have reserved a red and a green boat.
   iii. Find the sids of sailors with age over 20 who have not reserved a red boat. 8 L3 2 1

## Unit – II

4. a) Consider the following relation schema:
Employee (employee-name, street, city)
Works (employee-name, company-name, salary)
Company (company-name, city)
Manages (employee-name, manager-name)

Answer the following queries:
i. Find the names of all employees in the database who do not work for 'First Bank Corporation'. Assume that all people work for exactly one company.
ii. Find the names of all employees in the database who earn more than every employee of 'Small Bank Corporation'. Assume that all people work for at most one company. 8 L3 3 1,2

b) What is a view? Give the specification of view in SQL. 8 L1 4 1,2

5. a) Consider the Relation Schema R = (A, B, C, D, E, G) and the FD set
F = {AB → C, AC → B, AD → E, B → D, BC → A, E → G}.
Discover whether the following binary decomposition of R has the lossless join property with respect to F.
   R1 = {A, B, C} and R2 = {A, C, D, E,G}
   R1 = {A, B,C, D, E} and R2 = {A, D, G} 8 L3 4 1,2

b) How basic constraints are specified in SQL-99. Explain with example. Give syntax. 8 L1 3 1,2

6. a) Consider the following relations for an order processing database application in a company:

CUSTOMER (cust #: int, cname: string, city: string)
ORDER (order #: int, odate: date, cust #: int, ord-Amt: int)
ORDER – ITEM (order #: int, item #: int, qty: int)
ITEM (item #: int, unit price: int)
SHIPMENT (order #: int, warehouse#: int, ship-date: date)
WAREHOUSE (warehouse #: int, city: string)
Develop SQL queries for the following:
   i. Produce a listing: CUSTNAME, #oforders, AVG_ORDER_AMT, where the middle column is the total numbers of orders by the customer and the last column is the average order amount for that customer.
   ii. For each item that has more than two orders, list the item, number of orders that are shipped from at least two warehouses and total quantity of items shipped. 8 L3 3 1,2

| | | | | | | |
|---|---|---|---|---|---|---|
| | b) | What is the need for normalization? Explain first, second and third normal form by giving example for each. | 8 | L2 | 4 | 1,2 |

## Unit – III

| | | | | | | |
|---|---|---|---|---|---|---|
| 7. | a) | Explain Strict Two-Phase Locking protocol in detail. | 8 | L2 | 5 | 1 |
| | b) | Explain the following index data structures in detail:<br>i. Hash based indexing<br>ii. Tree based indexing | 8 | L2 | 5 | 1 |
| 8. | a) | Illustrate the anomalies associated with Interleaved Execution. Explain in detail. | 8 | L2 | 5 | 1 |
| | b) | Give the algorithm to insert an element in a B+ tree. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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